

Precept 4 Outline — Solutions

ECO 202 TAs

September 30, 2026

1 Weekly Check In

1. TAs will have grading done for Exam 1 soon.
2. Any questions about the class, comments?
3. Next exam is about two weeks away.
4. Reminders about resources:
 - Come to office hours for any preceptor, as much as you want.
 - Dexin: Tuesdays, 4-5pm, tables outside of JRR A19
 - Emily: Thursdays, 4:30-5:30pm, JRR A74
 - Ryan: Fridays, 2-3pm, JRR A74
 - Come to office hours of the professor: Fridays, 3pm-4pm, JRR 278
 - Check out the McGraw Center:
 - Individual tutoring: [Link to page](#)
 - Group study with a tutor: [Link to page](#)
 - Reach out to us by email if you've tried the above but are seeking support beyond these things.

2 Precept practice problems

1. (Practice 6, Question 2)

A finite population contains four manufacturing firms $M_1, \dots, M_4$ and four service firms $S_1, \dots, S_4$. Compare two designs that select four firms.

- **Design A:** Select an SRS of four firms from all eight.
 - **Design B:** Make one fair SRS selection of two manufacturing firms and a separate fair SRS selection of two service firms, with all $6 \times 6 = 36$ cross-sector combinations equally likely.
- (a) Calculate the number of possible samples under Design A and the probability of each possible sample.

Solution. Design A selects four of eight firms, so there are $\binom{8}{4} = 70$ possible samples. Because the design is an SRS, each four-firm subset has probability $1/70$.
 - (b) Calculate the number of possible samples under Design B and the probability of each allowed sample.

Solution. Design B selects two of four manufacturing firms and two of four service firms, so there are $\binom{4}{2}\binom{4}{2} = 6 \times 6 = 36$ possible samples. All 36 combinations are equally likely, so each allowed sample has probability $1/36$.

- (c) Calculate the inclusion probability of M_1 under each design. Explain why the same calculation applies to every firm.

Solution. Under Design A, M_1 is in every sample formed by choosing three of the other seven firms, so its inclusion probability is $\binom{7}{3}/\binom{8}{4} = 35/70 = 1/2$. Under Design B, M_1 is one of two firms selected from the four-firm manufacturing stratum, so its inclusion probability is $2/4 = 1/2$. By symmetry, the same calculation gives $1/2$ for every firm within each design: under A all eight firms are treated identically, and under B each stratum selects two of its four firms.

- (d) Find the probability of selecting the sample $\{M_1, M_2, S_1, S_2\}$ under each design.

Solution. This sample has two firms from each sector, so it is allowed under both designs. Its probability is $1/70$ under Design A and $1/36$ under Design B.

- (e) Find the probability of selecting $\{M_1, M_2, M_3, S_1\}$ under each design.

Solution. Under Design A it is one of the 70 equally likely subsets, so its probability is $1/70$. It violates Design B's two-per-sector rule, so its probability under Design B is 0.

- (f) Explain precisely why Design B is a probability sample and a stratified random sample but is not an SRS from the full eight-firm population, even though every firm has inclusion probability $1/2$.

Solution. Design B is a probability sample because its chance mechanism and the probability of every possible sample are known. It is stratified because selection happens separately within predetermined sectors. It is not an SRS from all eight firms because 34 of the 70 four-firm subsets (every 3–1 and 4–0 sector split) have probability zero, while the 36 allowed subsets have probability $1/36$ rather than $1/70$. Equal inclusion probabilities for individual firms do not make the probabilities of whole samples equal.

2. (Practice 6, Question 3)

A fictional firm wants the mean weekly overtime hours during 2025 among everyone who worked for it at any point that year. The firm uses its December 31 current-employee roster as the frame. Former employees, many of whom report leaving because of long hours, averaged 12 overtime hours per week, while current employees averaged 5. Among current employees, people who worked more than 8 overtime hours were twice as likely to complete the survey as other selected employees. Supervisors collected the answers directly, and follow-up evidence suggests that respondents tended to underreport overtime in those conversations.

- (a) For each stage—frame construction, response, and measurement—name the relevant bias mechanism and identify which units or values are disproportionately omitted or altered.

Solution. **Frame construction:** undercoverage. Former employees belong to the all-2025 target but have no path into the December 31 roster. **Response:** differential nonresponse. Selected current employees with more than 8 overtime hours are more likely to respond. **Measurement:** response bias. Respondents underreport overtime when supervisors collect the answers directly.

- (b) Under the stated facts, determine the direction in which the frame mismatch moves the current-frame mean relative to the target-population mean.

Solution. Downward. Former employees average 12 hours and current employees average 5, so excluding former employees makes the current-frame mean lower than the all-2025 target mean. The direction follows from the stated group means, not from the word “undercoverage” alone.

- (c) Determine the direction in which differential response moves the respondent mean relative to the current-frame mean.

Solution. Upward. Employees above 8 overtime hours have higher values by definition and are twice as likely to respond, so they are overrepresented among respondents relative to the current frame.

- (d) Determine the direction in which the stated response bias moves recorded reports relative to respondents' true overtime.

Solution. Downward. Respondents report fewer hours than they actually worked in supervisor-collected conversations, so the recorded mean is below the respondents' true mean.

- (e) Can the direction of the final reported mean relative to the target-population mean be determined from these facts alone? Explain why the signs of individual mechanisms do not supply their relative magnitudes.

Solution. No. Undercoverage and underreporting push the reported mean down, while differential response pushes it up. The facts give only the direction of each mechanism, not its size, so the net discrepancy cannot be determined by adding signs.

- (f) Explain why surveying every person on the current roster would remove one source of sampling variability but would not repair the target-population coverage or measurement problems.

Solution. A census of the roster removes the variability that comes from which rows are selected within that frame (and removes nonresponse only if everyone actually gives a usable answer). It still excludes former employees, who are part of the target, and it does not make supervisor-collected reports accurate. A census of the wrong frame with biased measurement can be precise about the wrong quantity.

3. (Practice 7, Question 2)

An outreach organization divides an eligible service area into four outreach zones, A–D. One experimental unit is one zone. The factor is the assigned outreach protocol, with levels **standard** and **enhanced**. The standard treatment uses the usual mailed notice; the enhanced treatment adds a follow-up phone call to the same notice. The outcome is the number of eligible households, out of five contacted in each zone, that schedule an enrollment appointment within one week. The following complete fictional table displays both potential outcomes only so that every causal quantity can be checked.

Unit	$Y_i(0)$	$Y_i(1)$
A	2	5
B	4	5
C	1	3
D	5	5

- (a) Identify the units, factor, its two levels, the two complete treatments, the control, and the outcome with its units. Let $D_i = 1$ denote enhanced outreach and $D_i = 0$ denote standard outreach.

Solution. **Units:** outreach zones A–D. **Factor:** assigned outreach protocol, with levels standard and enhanced. **Treatments:** standard = the usual mailed notice; enhanced = the same notice plus a follow-up phone call. **Control:** standard outreach. **Outcome:** the number of eligible households, out of five contacted in the zone, that schedule an enrollment appointment within one week, measured in scheduled appointments per zone.

- (b) Calculate every individual effect $Y_i(1) - Y_i(0)$ and the finite-sample average treatment effect

$$\text{ATE} = \frac{1}{4} \sum_{i=1}^4 [Y_i(1) - Y_i(0)].$$

Solution. The individual effects are A: $5 - 2 = 3$; B: $5 - 4 = 1$; C: $3 - 1 = 2$; D: $5 - 5 = 0$. Therefore $\text{ATE} = (3 + 1 + 2 + 0)/4 = 6/4 = 1.5$ appointments per zone.

- (c) Under Assignment I, zones A and B receive enhanced outreach. Construct all four observed outcomes and calculate

$$\hat{\tau} = \bar{Y}_{D=1} - \bar{Y}_{D=0}.$$

Solution. The observed outcomes are $(Y_A, Y_B, Y_C, Y_D) = (5, 5, 1, 5)$. Then $\bar{Y}_{D=1} = (5 + 5)/2 = 5$ and $\bar{Y}_{D=0} = (1 + 5)/2 = 3$, so $\hat{\tau} = 5 - 3 = 2$ appointments per zone.

- (d) Under Assignment II, zones C and D receive enhanced outreach. Repeat the observed-outcome construction and calculate $\hat{\tau}$.

Solution. The observed outcomes are $(Y_A, Y_B, Y_C, Y_D) = (2, 4, 3, 5)$. Then $\bar{Y}_{D=1} = (3 + 5)/2 = 4$ and $\bar{Y}_{D=0} = (2 + 4)/2 = 3$, so $\hat{\tau} = 4 - 3 = 1$ appointment per zone.

- (e) Explain why the potential outcomes and ATE are fixed across the two assignments while the observed outcomes and $\hat{\tau}$ change.

Solution. The table of $Y_i(0)$ and $Y_i(1)$ defines the unit effects and the ATE, and assignment does not change that table. Assignment only determines which of each zone's two potential outcomes is observed and which zones enter each group mean, so the observed outcomes and $\hat{\tau}$ vary with the assignment.

- (f) There are $\binom{4}{2} = 6$ equally likely assignments when exactly two of the four zones receive enhanced outreach. Enumerate the six possible enhanced-outreach pairs and calculate $\hat{\tau}$ for each. Verify that the estimates have mean 1.5 and range from -0.5 to 3.5 , then explain what each fact teaches about random assignment.

Solution.

Enhanced pair	AB	AC	AD	BC	BD	CD
$\hat{\tau}$	2	-0.5	2.5	0.5	3.5	1

The mean is $(2 - 0.5 + 2.5 + 0.5 + 3.5 + 1)/6 = 9/6 = 1.5$, which equals the ATE: averaging over the assignment mechanism recovers the causal target (Class 13 will call this unbiasedness). The range $[-0.5, 3.5]$ shows randomization variability: a single realized estimate need not equal the ATE, and it can even have the opposite sign from every (nonnegative) individual effect.

4. (Practice 7, Question 6)

A career center lets eligible applicants choose whether to attend a résumé workshop, then compares their earnings six months later. The attendees have higher mean earnings, and the center calls the difference the causal effect of the workshop.

- (a) Define the units, a sharply stated treatment or offer, D_i , $Y_i(1)$, $Y_i(0)$, and a finite-sample ATE.

Solution. One unit is one eligible applicant in the study pool. The treatment is attending the specified résumé workshop at a the stated time and format, or not; control does not experience the workshop. $D_i = 1$ if applicant i takes up the offer and $D_i = 0$ otherwise. $Y_i(1)$ is applicant i 's six-month earnings if they take-up the offer, and $Y_i(0)$ is the same applicant's earnings if they don't. For the N study applicants, $ATE = \frac{1}{N} \sum_{i=1}^N [Y_i(1) - Y_i(0)]$.

- (b) Explain why the self-selected comparison does not by itself estimate that ATE. Name two pre-treatment variables that could affect both workshop choice and later earnings.

Solution. Attendees chose to attend, so the two groups may differ before the workshop. The observed mean difference mixes any workshop effect with those pre-existing differences. Examples of pre-treatment variables that could affect both choice and earnings: motivation, prior earnings, work experience, education, job-search intensity, or access to other services (any two).

- (c) Repair the design using one of the following assignment mechanisms: complete randomization of a workshop offer, blocking by pre-treatment earnings, or matched pairs based on pre-treatment information. State the mechanism precisely, explain how the design provides replicated treated and control comparisons, and identify the causal contrast it supports.

Solution. **Complete randomization:** fix the eligible study pool before assignment, decide in advance how many offers are available (keeping multiple applicants in each condition), and choose the offer group uniformly at random. Many separately randomized applicants in each group provide replication. Compare mean earnings by *assigned* offer, keeping every applicant in their original assigned group whether or not they attend. This supports an average intention-to-treat effect of the offer for the randomized study pool. **Alternatives:** blocking defines pre-treatment earnings categories before assignment and randomizes offer vs. control within each category; matched pairs forms several pairs on pre-treatment variables and randomly offers the workshop to one member of each pair. Either can reduce baseline imbalance when the blocking or matching variables predict earnings, and the analysis must keep the blocks or pairs.

- (d) State separately whether the applicant pool was obtained through probability sampling and what population the design directly describes.

Solution. No probability-sampling mechanism is stated for the applicant pool. Random assignment supports the causal comparison within that pool only; it does not show that the pool represents all career-center clients, all job seekers, or any other population.

- (e) Name two important limitations that could remain after valid random assignment.

Solution. Any two of: noncompliance with the offer; missing six-month outcomes (attrition); spillovers between applicants (e.g., sharing materials); varying workshop delivery; earnings measurement error; limited applicability beyond the study pool.

- (f) Write a two-sentence conclusion template that an analyst could complete after observing the assignment-based mean difference, without claiming a receipt effect, an effect for every applicant, or automatic generalization.

Solution. "Among the eligible applicants randomized in this study, mean six-month earnings were [amount and units] [higher/lower] for applicants assigned an offer of the specified workshop than for applicants assigned to control. Under the implemented random assignment, this is an average effect of the offer for the study group; it is not automatically an effect of attendance for every applicant or a result for a broader population."

3 Extra practice problems

1. In the context of sampling, what is a “parameter”?

Solution. A parameter is a numerical measure of the population distribution (for example, the population mean).

2. What is a probability sample?

Solution. A probability sample is a sample chosen by chance, where we know which samples are possible and what chance (probability) each possible sample has.

3. What, precisely, is a simple random sample?

Solution. A simple random sample of size n from a finite population is a sample chosen so that every set of n distinct individuals in the population has the same chance of being the sample selected. (Equal inclusion probabilities for individuals follow from this, but they are not enough to define an SRS.)

4. Give an example of a “parameter” and of a “statistic”.

Solution. Suppose we take a sample of Americans and measure their heights. The mean height in the sample is a statistic. The mean height of all Americans (the population mean) is a parameter, which we might want to learn about using the sample mean.

5. Suppose a population consists of 5 individuals, and we draw a Simple Random Sample of size 2. How many possible samples are there, and what is the probability that each is chosen as the sample?

Solution. Call the individuals A, B, C, D, E. There are $\binom{5}{2} = 10$ possible samples: $\{A, B\}$, $\{A, C\}$, $\{A, D\}$, $\{A, E\}$, $\{B, C\}$, $\{B, D\}$, $\{B, E\}$, $\{C, D\}$, $\{C, E\}$, $\{D, E\}$. In an SRS each is equally likely, so each has probability $1/10$.

6. Consider a population that consists of 5 individuals A, B, C, D, E. Construct a probability sample for a sample size $n = 3$ such that all individuals are equally likely to be in the sample, but the design is not a simple random sample.

Solution. One answer: pick one of the following five sets with equal probability $1/5$: $\{A, B, C\}$, $\{B, C, D\}$, $\{C, D, E\}$, $\{D, E, A\}$, $\{E, A, B\}$. Each individual appears in exactly three of the five sets, so each has inclusion probability $3/5$. It is not an SRS because many three-person sets, such as $\{A, B, D\}$, can never be chosen.

7. Consider a simple random sample of size $n = 2$ from a population of 4 individuals A, B, C, D.

- (a) What are the possible samples, and the associated probabilities?

Solution. The possible samples are $\{A, B\}$, $\{A, C\}$, $\{A, D\}$, $\{B, C\}$, $\{B, D\}$, $\{C, D\}$, each with probability $1/6$.

- (b) Is the event “A or B, or both, are in the sample” independent of the event “C is not in the sample”?

Solution. No. Only $\{C, D\}$ contains neither A nor B, so $P(\text{A or B in sample}) = 5/6$. If C is not in the sample, the sample is $\{A, B\}$, $\{A, D\}$, or $\{B, D\}$, each of which contains A or B, so $P(\text{A or B in sample} \mid \text{C not in sample}) = 1 \neq 5/6$. The events are not independent.

- (c) Suppose we learn that A is in the sample. What is the conditional probability that also B is in the sample?

Solution. $P(B \in \text{sample} \mid A \in \text{sample}) = \frac{P(A \text{ and } B \in \text{sample})}{P(A \in \text{sample})} = \frac{1/6}{1/2} = \frac{1}{3}$.

8. Consider a population of 4 individuals A, B, C, D. A, B are male, and C, D are female. Consider a stratified random sample of size $n = 2$, with strata defined by gender. What samples are possible, and what are their probabilities?

Solution. One individual is drawn from each stratum, so the possible samples are $\{A, C\}$, $\{A, D\}$, $\{B, C\}$, $\{B, D\}$, each with probability $1/4$.

9. Consider a population of four individuals A, B, C, D. Describe a random sample design of size $n = 2$ so that each individual has the same chance to be in the sample, but the design is not a simple random sample. Explain.

Solution. One example: first pick A or B with probability $1/2$ each, then pick C or D with probability $1/2$ each. All four individuals have probability $1/2$ of being in the sample, but this is not an SRS because, for example, A and B can never appear together.

10. A population consists of 5 individuals A, B, C, D, E. Design a probability sample of size $n = 3$ such that each individual is selected with the same probability, but the sample design is not a simple random sample.

Solution. There are many correct answers. One: draw one of the triplets $\{C, D, E\}$, $\{A, D, E\}$, $\{A, B, E\}$, $\{A, B, C\}$, $\{B, C, D\}$ at random, each with probability $1/5$. Each individual has a $3/5$ chance of being drawn, but this is not an SRS because, for example, $\{A, C, D\}$ can never be selected.

11. Consider a population of five individuals A, B, C, D, E. Suppose we take a simple random sample of size $n = 4$. What is the probability that both A and B are in the sample?

Solution. Each sample of four leaves out exactly one individual, so there are only 5 possible samples, all equally likely. Three of them contain both A and B: $\{A, B, C, D\}$, $\{A, B, C, E\}$, $\{A, B, D, E\}$. So the probability is $3/5$.

12. Consider the first paragraph of the Class of 2025 pre-read *Moving Up Without Losing Your Way* below. In particular, do you think the literal interpretation of the sentence “If you had been born in San Jose, California, instead, your chances would have increased threefold” is warranted? Discuss using relevant jargon, and provide a reformulation of the sentence that is not subject to the same criticism.

Each of us starts off in a particular position in society. That position is determined by where and when you are born, the socioeconomic and educational level of your parents, your race, and your gender, among other things. Social scientists have produced vast amounts of evidence showing that these factors play a significant role in determining life prospects. In the United States, if you were born in the bottom fifth of the income scale in Charlotte, North Carolina, between 1980 and 1982, your chances of making it to the top fifth were 4.4 percent. If you had been born in San Jose, California, instead, your chances would have increased threefold.

Solution. The literal interpretation is not warranted. Read literally, the sentence claims a causal effect of being born in San Jose (instead of Charlotte) on the chance of reaching the top fifth of the income distribution. But the comparison is observational, and many omitted or confounding variables that are correlated with location, such as income, race, and educational opportunities, also affect these chances. A child born under exactly the same circumstances might have the same chance in either city; the circumstances themselves differ across cities. A better phrasing: “Due to its different socioeconomic, educational, racial (etc.) composition, the average child born in San Jose has three times the chance of making it to the top fifth of the income distribution as the average child born in Charlotte.”

13. Give an example of a “matched pair design”, and explain its appeal.

Solution. One example is using data on twins to study the return to education. The appeal is that twins in the same family have approximately the same genetic and family characteristics, which are otherwise hard to measure, so comparing within pairs removes those differences from the comparison.

14. Observational studies suggest that countries with more generous unemployment benefits have higher rates of unemployment. If countries were to make unemployment benefits less generous, would this lead to lower unemployment rates? Discuss using relevant jargon.

Solution. We cannot say for certain. The observational evidence shows a positive association between benefit generosity and unemployment rates, but association does not imply that making benefits less generous would cause unemployment to fall. Omitted (lurking) variables could explain both. For example, a country whose labor force works mostly in highly volatile industries might both have high unemployment and pass generous unemployment-benefit laws.